Year 11 Mathematics

Handling Data 4

4.4 Misleading Data Presentation

Misleading Data Presentation

STARTER · 5 MIN

Date: _____ **Title:** Misleading Data Presentation

LEARNING OBJECTIVES

- Spot misleading scales and axes
- Spot misleading pictures and missing information
- Explain why a graph is misleading and how to fix it

STARTER *Answer in your book. No calculators.*

1. Work out 55 − 52
2. Work out 3 ÷ 52 × 100 (1 d.p.)
3. Square sides double. Area is × ?
4. Work out 40 + 35 + 30

Starter — Answers

STARTER

Tick or correct in green pen

1. **3**

2. **5.8%**

3. $2 \times 2 = \mathbf{4}$

4. **105**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can spot misleading scales and axes	☐
SILVER	I can spot misleading pictures and missing information	☐
GOLD	I can explain why a graph is misleading and how to fix it	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Misleading Graphs

INTRODUCTION · 3 MIN

Graphs can be drawn to give the wrong impression. Always check the scales and labels.

Axis not from 0

Small differences look big.

Vertical scale starts at 50

Uneven scale

Gaps between numbers are not equal.

0, 5, 10, 50, 100

Missing labels

No title, axis labels or key.

What is being measured?

Pictures

Bigger pictures exaggerate.

Double the size = 4 times the area

Examples 1 and 2 — Misleading Scales

EXAMPLES · 5 MIN

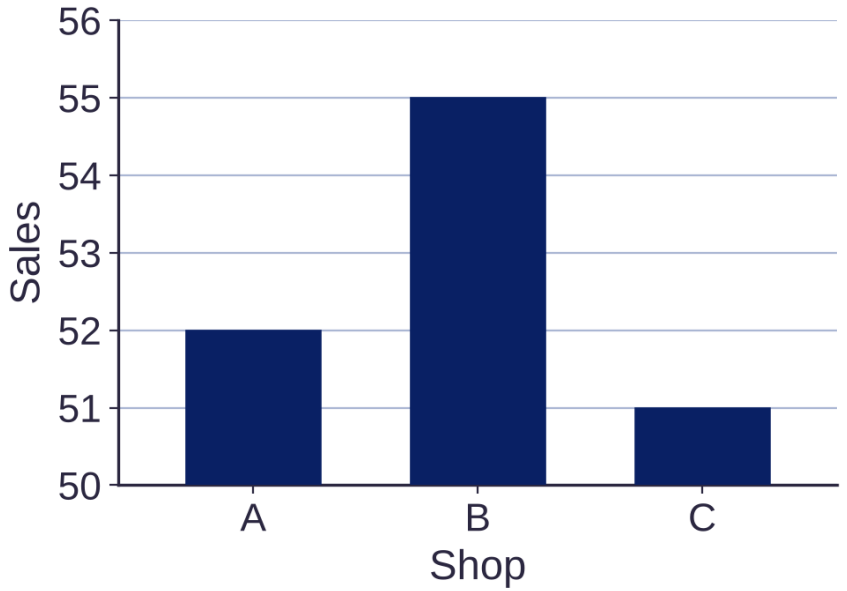
BRONZE

Watch and copy into your book

Hint $\%\ \text{change} = \frac{\text{change}}{\text{original}} \times 100$

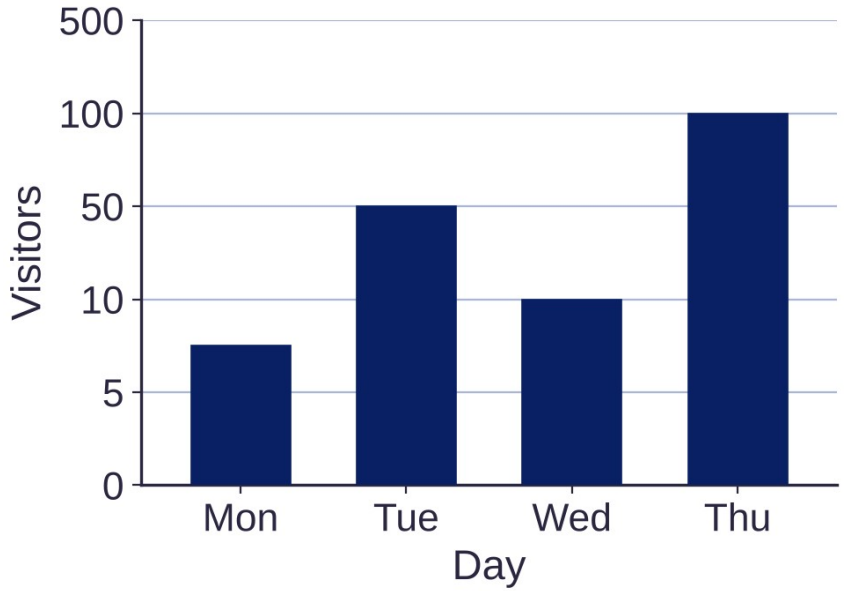
Example 1

Why is this bar chart misleading?



Example 2

Why is this bar chart misleading?



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

The vertical axis starts at 50

B is only 3 more than A

B looks much bigger than it is

Tip: always check where the axis starts.

Example 2

0, 5, 10, 50, 100, 500

The scale is not even

Tip: read every number on the scale.

Task 1

PRACTICE · 9 MIN

BRONZE *Use the bar chart*

Hint $\%\ \text{change} = \dfrac{\text{change}}{\text{original}} \times 100$

1. Where does the vertical axis start?

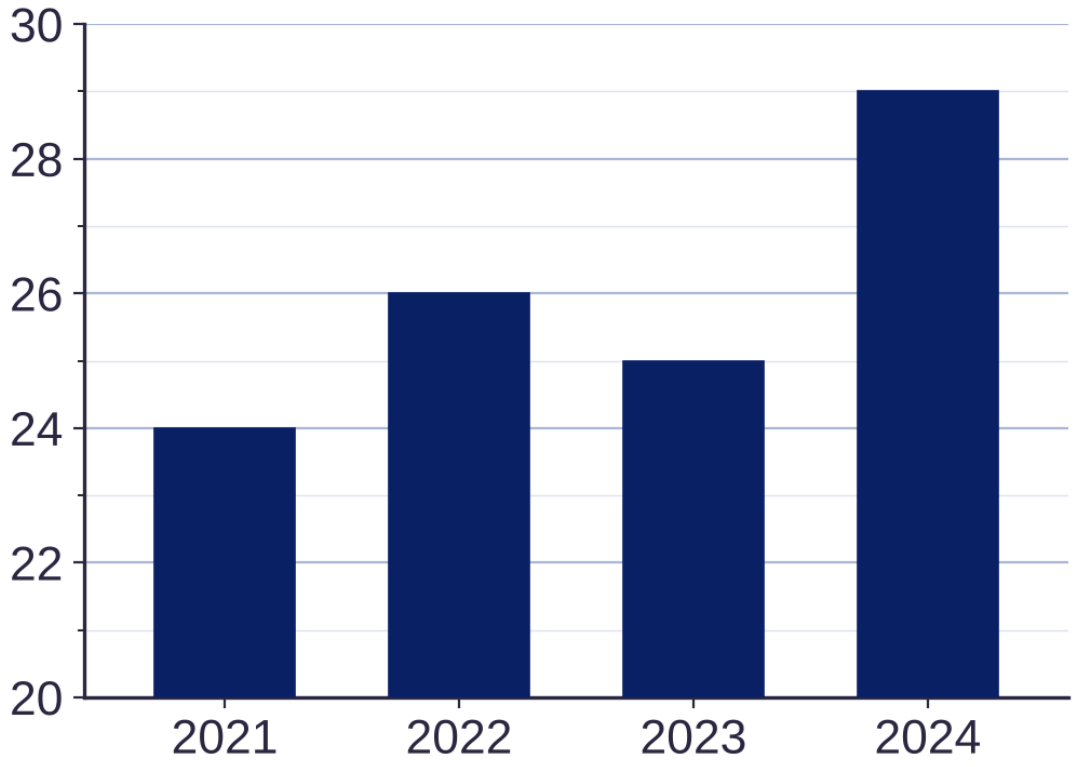
2. Read the value for 2024

3. Read the value for 2021

4. Find the real increase from 2021 to 2024

5. Write down one piece of missing information

★ 2024 looks over twice 2021. How many times bigger is it really? (1 d.p.)



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **20**

2. **29**

3. **24**

4. $29 - 24 = \mathbf{5}$

5. **No axis labels or title**

★ $29 \div 24 = \mathbf{1.2\ times}$

Examples 3 and 4 — Misleading Pictures and Information

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Example 3

A picture for 2024 is twice as tall and twice as wide as for 2023. Why is this misleading?

Example 4

A pie chart of votes shows: A 50%, B 30%, C 25%. What is wrong?

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Area: 2 × 2 = 4

Looks 4 times as big

Tip: area grows faster than length.

Example 4

50 + 30 + 25 = 105%

It must total 100%

Tip: pie chart percentages add to 100%.

Task 2

PRACTICE · 9 MIN

SILVER

Answer in your book

1. A graph has no title. Why is this a problem?

2. Pie chart percentages: 40%, 35%, 30%. What is wrong?

3. Bars of different widths are used. Why is this misleading?

4. A picture is 3 times as tall and 3 times as wide. How many times bigger does it look?

5. A chart shows only 2 months of data. Why may it mislead?

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **We do not know what it shows**

2. $40 + 35 + 30 =$ **105%**

3. **Wider bars look bigger**

4. $3 \times 3 =$ **9 times**

5. **Too little data to show a trend**

Examples 5 and 6 — Explaining and Correcting

EXAMPLES · 5 MIN

GOLD

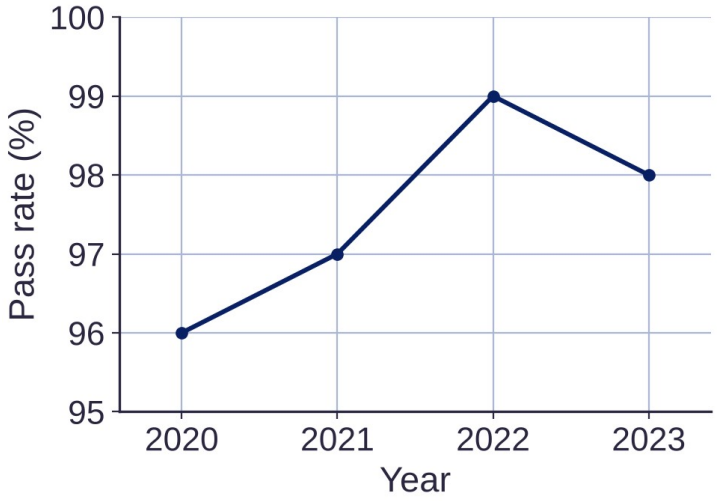
Exam-style: show all your working

Hint $\%\ \text{change} = \dfrac{\text{change}}{\text{original}} \times 100$

Example 5

(2 marks)

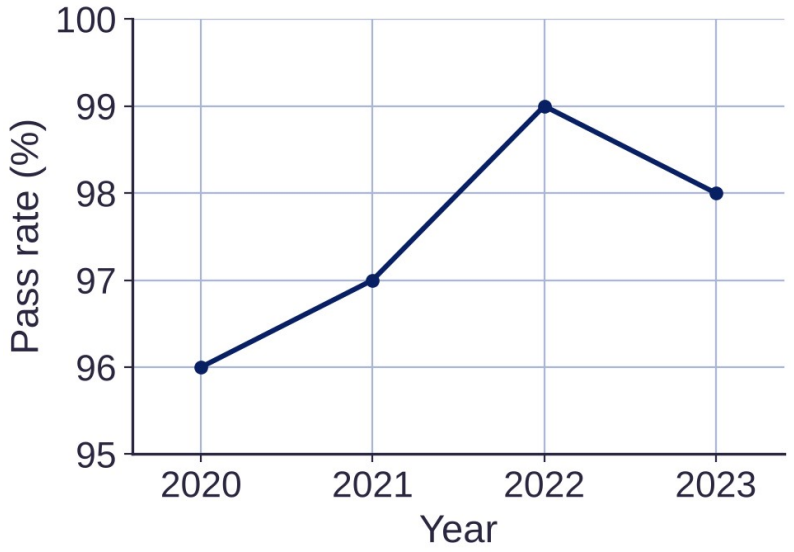
A school says results rose hugely. Give two reasons the graph is misleading.



Example 6

(2 marks)

How should the graph be redrawn so it is fair?



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

The axis starts at 95%, not 0

The rise is only 3%

A small change looks big

Tip: give the real numbers.

Example 6

Start the vertical axis at 0

Use an even, labelled scale

Tip: say exactly what to change.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $\%\ \text{change} = \frac{\text{change}}{\text{original}} \times 100$

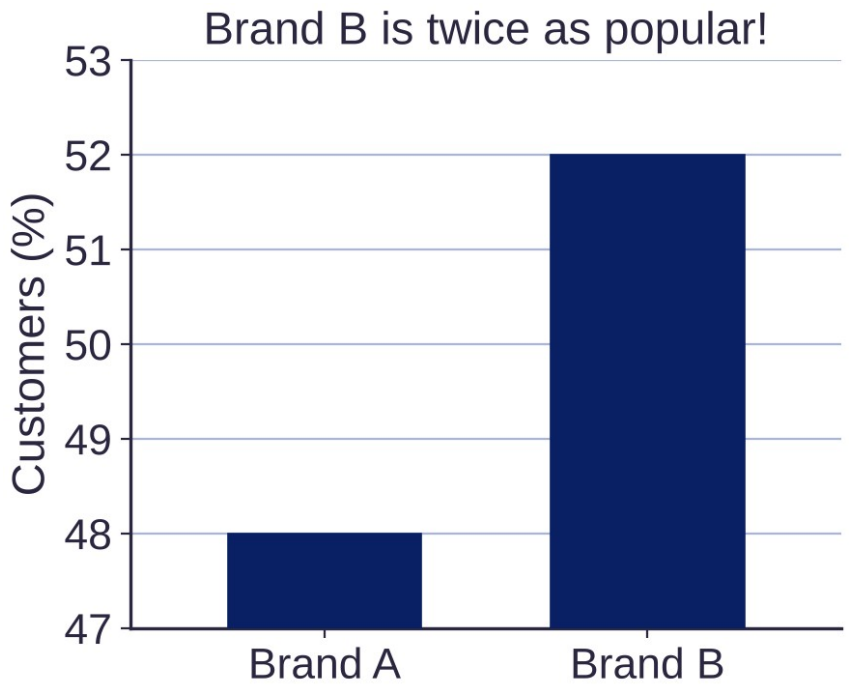
1. Write down the percentage for each brand. (1 mark)

2. Is Brand B twice as popular? Show working. (2 marks)

3. Explain why the chart is misleading. (1 mark)

4. Where should the vertical axis start? (1 mark)

5. The difference is what percentage of Brand A? (1 d.p.) (2 marks)



Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. A 48%, **B 52%**
2. $52 \div 48 = 1.08$, **No**
3. **The axis starts at 47%**
4. **0%**
5. $4 \div 48 \times 100 =$ **8.3%**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE A bar chart's axis starts at 90. Why can this mislead?

SILVER Pie chart: 50%, 30%, 30%. What is wrong?

GOLD A picture doubles in length and width. How many times bigger does it look?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **Small differences look big**

SILVER $50 + 30 + 30 =$ **110%**

GOLD $2 \times 2 =$ **4 times**

Exam Practice — Misleading Scales

EXAM PRACTICE · 10 MIN

BRONZE

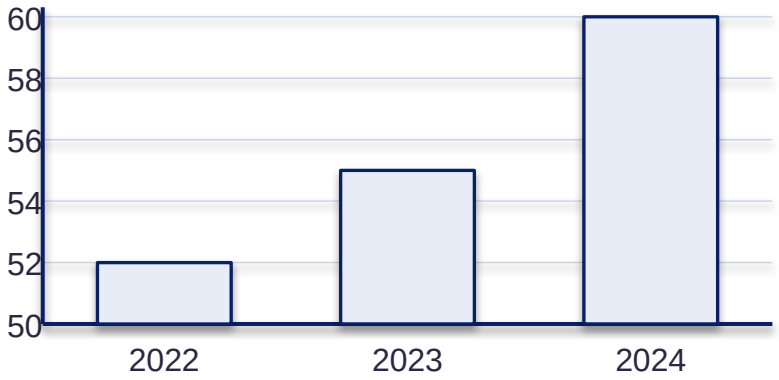
IGCSE exam-style: show all your working

Hint $\%\ \text{change} = \dfrac{\text{change}}{\text{original}} \times 100$

Question 1

(2 marks)

The bar chart shows sales.



(a) Give two reasons why the bar chart is misleading. (2)

.....

Question 2

(3 marks)

The vertical axis of the chart in Question 1 starts at 50.

(a) The 2024 bar looks 5 times as tall as the 2022 bar. Explain why. (1)

.....

(b) Work out the real percentage increase from 2022 to 2024. Give your answer to 1 d.p. (2)

..... %

Exam Practice — Misleading Pictures and Claims

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: answer on the line, show all your working

Question 3

(2 marks)

Pictures of rice bags show sales.

The 2024 picture is twice as tall and twice as wide as the 2023 picture.

(a) How many times bigger does the 2024 picture look? (1)

.....

(b) Sales doubled. Explain why the pictures are misleading. (1)

.....

Question 4

(2 marks)

An advert says '80% of people prefer Brand X'.

Only 10 people were asked, all outside a Brand X shop.

Give two reasons why the claim may be misleading.

.....

Exam Practice — Explaining and Correcting

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $\%\ \text{change} = \frac{\text{change}}{\text{original}} \times 100$

Question 5

(2 marks)

A graph's vertical axis is labelled 0, 10, 20, 50, 100 with equal gaps.

(a) What is wrong with the scale? (1)

.....

(b) How does this change the look of the data? (1)

.....

Question 6

(3 marks)

Shop A's sales: £40 000 to £44 000

Shop B's sales: £10 000 to £12 500

(a) Work out each percentage increase. (2)

(b) Which shop improved more? (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

axis does not start at 0 — B1

no title or axis label — B1

Tip: always check where the scale starts.

Question 2

(a) 52 − 50 = 2, 60 − 50 = 10 — B1

(b) 60 − 52 = 8 — M1

$\frac{8}{52} \times 100 = 15.4\%$ — A1

Tip: the real increase is small.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $2 \times 2 = 4$ times — B1

(b) the area is 4 times, not 2 — B1

Tip: double each length = 4 times the area.

Question 4

sample too small (10 people) — B1

biased: all outside Brand X — B1

Tip: ask who was asked and how many.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) the scale is not even B1

(b) big values look closer B1

Tip: equal gaps need equal steps.

Question 6

(a) A: $\frac{4000}{40\,000} \times 100 = 10\%$ M1

B: $\frac{2500}{10\,000} \times 100 = 25\%$ A1

(b) Shop B: 25% > 10% B1

Tip: compare percentages, not amounts.